Supervised Alignment Perception and Modality Difference Selection Enhanced SAR-Optical Fusion for Cloud Removal

Kai Xu, *Member, IEEE*, Anling Wang, Zhou Zhu, Wei Ju, and Wenchao Huang

***Abstract*—Leveraging synthetic aperture radar (SAR) data to assist in cloud removal demonstrates promising potential and has achieved satisfactory results in restoring clear contours of images. However, there are three critical challenges: inherent geometric discrepancies between SAR and optical images complicate cross-modal correlation modeling; modality-specific noises lead to radiometric inconsistency and texture loss in cross-modal information aggregation; and diverse clouds and complex scenes impede the attainment of both detailed inpainting and global consistency. Therefore, we propose a novel network integrating supervised alignment perception strategy (SAPS) and modality difference selection module (MDSM) to enhance SAR-optical fusion for cloud removal. Specifically, the SAPS consists of supervised training and an alignment perception module (APM) to address spatial misalignments and enhance the stability of deviation perception. Then, the MDSM dynamically focuses on different regions and eliminates modality-specific noise through mutual dynamic guidance. Furthermore, the locality-enhanced Mamba (LEM) module in reconstruction optimization enhances the consistency of the overall reconstruction results and the expressiveness of local details. Experiments conducted on the SEN12MS-CR, LuojiaSET-OSFCR, and simulated WHU-OPT-SAR datasets assess reconstruction performance from multiple perspectives. The proposed method not only achieves superiority in basic quantitative and qualitative validation but also demonstrates advantages in reconstruction under different cloud coverage levels and across various terrains.**

***Index Terms*—Cloud removal, data fusion, mamba, multiscale representation, polynomial transformation model, synthetic aperture radar (SAR).**

I. Introduction

OPTICAL remote sensing imagery serves as a cornerstone of Earth observation, offering periodic and large-scale surface information essential for a wide array of applications [1]. However, atmospheric cloud occlusion significantly disrupts surface reflection signals received by optical sensors, resulting in unobservable regions and temporal discontinuities in the imagery. These limitations adversely affect downstream tasks such as change detection, agricultural monitoring, and disaster risk assessment [2].

To enhance the quality of optical imagery, cloud removal techniques have evolved along multiple technical dimensions, enabling adaptation to diverse application scenarios such as thin cloud correction, temporal data interpolation, and cross-sensor fusion in persistently cloudy regions [3]. Among these, synthetic aperture radar (SAR) has emerged as a pivotal enabler for cross-sensor fusion, primarily due to its unique all-weather imaging and penetrating capabilities [4], [5]. Furthermore, its sensitivity to roughness and geometry of the ground surface renders it uniquely suited for geometric constraint in optical image restoration. Therefore, surface heterogeneity and geometric constraints provided by SAR guide more physically and radiometrically accurate recovery of cloud-obscured information [6], [7].

Although SAR can compensate for the structural details of ground objects in cloud-covered areas, effective fusion methods rely on the strict registration of the two modalities. However, owing to distinct imaging principles, differences in geometric characteristics, and the complexity of terrain undulations, partial misalignments may remain in SAR and optical imagery even after registration [8]. Most existing methods exhibit suboptimality in SAR-based compensation for cloudy regions due to residual spatial misalignments that introduce radiometric artifacts and geometric inconsistencies [9]. To tackle spatial misalignments, Align-CR [10] incorporates deformable convolution (DConv) [11] before fusion and rectifies geometric discrepancies in a pyramid structure. However, this implicit offset learning approach exhibits instability, potentially causing excessive bias suppression and compromising the useful information in SAR. Therefore, addressing spatial misalignment and enhancing the stability of deviation perception are crucial for fusion improvement.

Furthermore, given the inherent modality-specific noise in both modalities, such as speckle noise in SAR and diverse clouds in optical images, the complexity of cloud removal is significantly increased. Specifically, these degradations disrupt effective cross-modal information extraction during fusion, resulting in radiometric distortions and texture loss in reconstructed regions. To mitigate noise amplification during simple pixel-level fusion, existing approaches typically employ dynamic convolution [12] and attention weighting [13] to

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suppress the impact of speckle noise in SAR. In addition, when handling variable clouds, current methods emphasize discrimination between cloudy and cloud-free regions to avoid interference from invalid cloud pixels. Specifically, USSRN-CR [14] utilizes gated convolution during optical feature extraction to mitigate the damage caused by invalid pixels to valid features in optical images. In contrast, SCT-CR [15] only uses four convolutional layers to extract spatial features from the input cloudy image and generate a cloud attention map. However, these methods lack mutual guidance between modalities during feature fusion, only performing information filtering within individual modalities. Considering that these isolated strategies are insufficient for hybrid degradation scenarios, it is necessary to address modality-specific degradation through mutual dynamic guidance.

To address these issues, we propose SADS-CR, a novel network emphasizing spatial alignment and difference selection in SAR and optical fusion for cloud removal. The framework is structured to integrate three core components, each addressing specific limitations in existing methods. First, to establish a robust spatial matching relationship at the feature level, we propose a supervised alignment perception strategy (SAPS). This strategy consists of supervised training that introduces spatial biases into a polynomial transformation model, and an alignment perception module (APM) that integrates the flow-guided and deformable alignment approaches. Subsequently, the modality difference selection module (MDSM) dynamically prioritizes modality-complementary features through a 3-D assessment of corrected SAR and optical features. This module adaptively filters redundant information and noise, ensuring that the fusion process emphasizes the complementary strengths of SAR and optical imagery in cloud-affected regions. Finally, state space models (SSMs) [16] are embedded into the reconstruction optimization design. By combining bidirectional scanning with the Hilbert curve, this design leverages long-range context information and enhances local correlations to facilitate the reconstruction of cloud-free images. The components described above are intrinsically interrelated, forming a synergistic pipeline. Spatial alignment, enabled by SAPS as a foundational step, resolves cross-modality geometric misalignment, ensuring that SAR-optical feature pairs are semantically correspondent. Upon this alignment, MDSM filters redundant information and noise from the aligned features. By leveraging the high-quality fused features from MDSM, the reconstruction optimization integrates long-range contextual dependencies and local feature fidelity to yield clearer ground-object boundaries and globally consistent spectral information reconstruction. Consequently, our main contributions can be summarized as follows.

1) We propose a cloud removal network that integrates spatial alignment perception and modality difference selection in SAR-optical fusion, as well as global–local feature optimization in reconstruction.
2) To facilitate the establishment of an accurate spatial matching relationship between the two modalities, we propose a supervision strategy that emphasizes the introduction of manual offsets in spatial bias rectification and employs a superimposed alignment structure to eliminate the bias of SAR relative to optical features.
3) To filter out irrelevant information and focus on the parts that can reflect the unique values of different modalities, we propose the MDSM. This module can dynamically concentrate on different regions with missing information and eliminate noise interference.
4) To strengthen the consistency of the overall reconstruction results and the expressiveness of local details, we adopt the SSMs that combine bidirectional scanning and the Hilbert curve in the cross-modal feature optimization design.

The rest of this study is structured in the following manner. Section II reviews the related work on cloud removal. A detailed description of the proposed method is presented in Section III. The results and analysis of the experiments are presented in Section IV. Finally, Section V offers a conclusion to this article.

II. Related Work

A. Cloud Removal

Cloud removal has emerged as a critical preprocessing step for enhancing the quality of optical satellite imagery. In recent decades, continuous advancements have expanded application scenarios, characterized by two key evolutionary trajectories.

1) Data Dimension Expansion: With advancements in remote sensing satellite technology, the availability of Earth observation data has been substantially increased. To enhance the recovery of optical imagery in complex scenarios, the data employed have evolved from single-scene to temporal sequence and from single-sensor to multisensor acquisitions.

For single-scene data, the process methods are simple, primarily relying on spatial similarity and spectral correlation for information recovery [17]. Xu et al. [18] estimated the relationship between visible bands and the cirrus band in homogeneous land cover areas through linear regression to assist in separating thin clouds from the background. Chen et al. [19] proposed a neighboring similar pixel interpolator approach, which calculates weights of similar pixels and recombines them to derive predicted information of ground objects under clouds. However, they can only mitigate minor data loss caused by thin clouds or reconstruct relatively simple scenarios. With the integration of temporal data and the performance of spatiotemporal interpolation using cloud-free optical imagery from adjacent time points, the more severe cloud contamination can be addressed. Chen et al. [20] adopted a spatial and temporal weighted regression model to optimally integrate complementary information from invariant similar pixels. Lin et al. [21] leveraged coupled tensor decomposition to explore relationships between multitemporal images and reconstruct multitemporal information based on the augmented Lagrange multiplier method. Nevertheless, temporal data acquisition fails to meet the rapid response requirements of time-sensitive tasks [22]. To overcome the infeasibility of acquiring cloud-free optical imagery within operational timeframes, research on multisensor data has also attracted considerable attention. Notably, SAR data with high structural fidelity and fine spatial resolution offer a viable alternative for maintaining data continuity [23]. However, this integration faces critical challenges, including the inherent noise in each modality and substantial differences between modalities.

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2) Methodological Paradigm Shifts: With the maturation of deep learning frameworks, the field of low-level vision has seen significant breakthroughs. These advancements indirectly facilitate cloud removal in optical remote sensing. In recent years, a surge of work has shifted from model-based to data-driven methods.

Early cloud removal work relied on specific models and priors to construct cloud-free information search strategies. For example, MDCP [24] enhances the application of the dark channel prior model in thin cloud removal by first separating high- and low-frequency information via multiscale transformation, exclusively integrating the latter into low-frequency components. CR-NAPCT [25] removes thin clouds by leveraging the correlation between the signal-to-noise ratio and spatial correlation and removing the cloud component from the first NAPCT component.

Unlike model-based methods, data-driven deep learning frameworks excel in automatic feature learning [26]. Recent advancements in convolutional neural networks (CNNs) have led to notable improvements in cloud removal performance. Specifically, end-to-end frameworks leverage deep feature hierarchies to transform cloud-contaminated optical imagery into cloud-free products [27], [28]. However, CNNs with inherently localized receptive fields and static weight configurations fail to satisfy the global consistency demands of image reconstruction. To address this limitation, transformers [29], [30] have been widely employed to exploit their superior capacity for comprehensive contextual aggregation. For example, CloudTrans [31] proposes a multitemporal cloud removal method based on axial transformer networks, which reduces computational load while maintaining a global receptive field. HPN-CR [32] designs a heterogeneous parallel network, respectively, adopting CNN and transformer branches to enhance local details in SAR and overall consistency in optical images.

Recently, SSMs have surpassed transformers in handling a wide range of natural language sequences and have shown promising applications in image processing [33]. Except for the generic framework Vision Mamba [34] and Vmamba [35], its superiority has been demonstrated in low-level version tasks. Specifically, MambaIR [36] in super-resolution and RainMamba [37] in deraining are prime examples. Transferring the success of using state-space models for image restoration to the application in cloud removal can achieve global optimization while alleviating computational burden.

Inspired by the above discussion, we fully explored the advantages at the data and technical levels in cloud removal applications by integrating SAR data as guidance input and the Mamba structure as the feature refinement design.

B. Sar-Assisted Cloud Removal

To achieve cloud removal, an SAR-assisted method usually extracts compensated information from SAR data and applies it to fill the corresponding cloud-obscured areas. Early methods adopted the simple fusion design, such as concatenation in the channel dimension [38], and incorporated additional prior knowledge [39] to constrain the fusion content. However, these methods only considered the consistency between the two modalities while neglecting inherent disparities and the influence of nonideal noise conditions. In response to this, some methods centered on mitigating noise sources in the two modalities to facilitate reconstruction quality. Xu et al. [12] incorporated dynamic convolution into the SAR feature extraction branch to suppress speckle noise. HS2P [13] adopts the channel attention mechanism, adaptively emphasizing more informative features in SAR-optical fusion. HDRSA-Net [40] utilizes a dynamic mapping of omni-dimensional features to suppress speckle noise interference in SAR images.

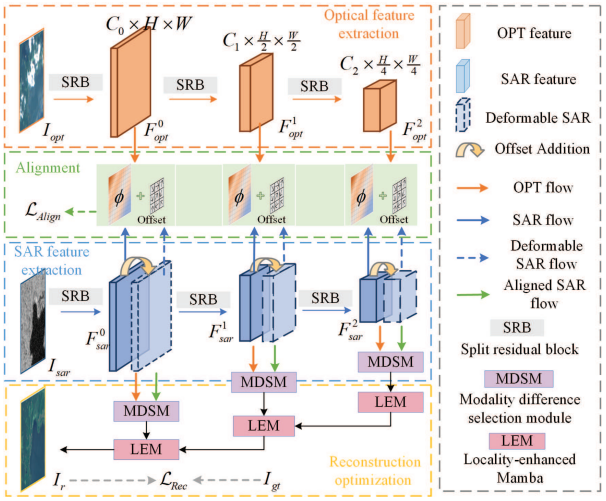


Fig. 1. Overall structure employs a U-shaped architecture. Two branches in the encoder are used, respectively, to extract features. Then, SAPS is used to rectify spatial biases existing in SAR, and the MDSM is employed to compensate for the missing information in the optical images. Finally, the decoder optimizes fused features to obtain the reconstructed images.

To address interference from cloudy pixels, current methods primarily employ missing region localization strategies, including separate and embedded architectures. In separate architectures, cloud masks are typically binary maps generated by cloud detection algorithms, which provide pixel-level localization but lack spatial context. For example, Dsen2-CR [38] designs a cloud-adaptive regularized loss to emphasize pixel reconstruction exclusively in cloudy areas. TransFusion-CR [41] designs a cloud mask score-guided feature fusion module, which introduces CMS-guided loss to optimize the training process. However, the method of obtaining this mask score is independent of the cloud removal network. Embedded architectures usually provide confidence guidance via cloudy probabilistic maps. Xiang et al. [42] designed a region-gated module to refine the restored information by treating features in cloudy and noncloud regions differently. However, these approaches only impose mask guidance on cloudy optical images and fail to emphasize the use of spatial confidence maps for guiding the fusion process. SCT-CR [15] enhances cross-modal fusion by incorporating cloud probability maps into synergistic convolution, improving generalization across cloud conditions. However, its single-scale spatial confidence prediction ignores spatial response variations in multiscale feature maps across modalities, impeding adaptive scale-wise information adjustments.

Building upon the discussed noise interference mitigation strategies, our fusion design retains the advantages of

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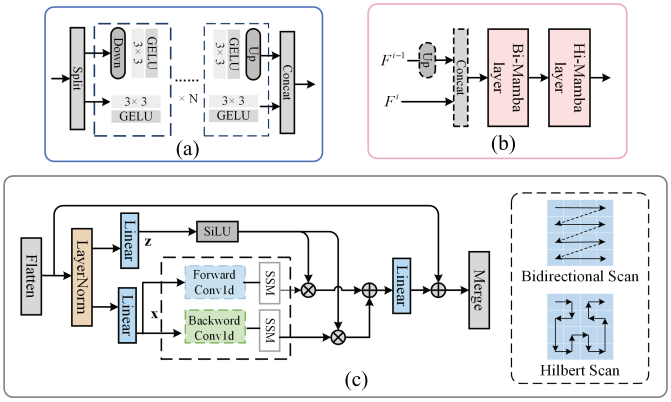


Fig. 2. (a) Details of SRB. (b) Details of LEM. (c) Details of the Mamba layer and two scan mechanisms (bidirectional scan and Hilbert scan).

feature selection and noise filtering. Furthermore, a differential approach is employed to enhance discriminative information extraction, enabling dynamic adjustment of available SAR information filling based on states of cloudy regions.

III. Methods

A. Overall of the Proposed Network

As shown in Fig. 1, the inputs from two sources are denoted as I_{opt} and I_{sar}, which have the same height (H) and width (W). In U-shaped multiscale structure, $\{F^i_{\text{sar}}, F^i_{\text{opt}}\}_{i=0,1,2}$ are obtained in encoder branch consisting of split residual block (SRB). The superscript i represents the features obtained in the ith stage with a size of $H/2^i$ and $W/2^i$. Then, the SAPS and the MDSM will process the obtained features of each scale to obtain the fused result F^i_f. In the decoder stage, the locality-enhanced mamba (LEM) conducts global optimization and multiscale feature integration. Finally, the reconstructed cloud-free image I_{rec} is achieved. Through the integration of the merits of the aforementioned modules, the aligned features enable effective modality selection, which in turn provides robust inputs for context-aware reconstruction.

B. U-Shaped Multiscale Structure

First, the two branches in the encoder are independent of the interaction between the two modalities during the feature extraction. Three SRBs and two-stage downsampling convolutions are adopted to generate $\{F^i_{\text{sar}}, F^i_{\text{opt}}\}_{i=0,1,2}$. By evenly splitting the input features along the channel dimension, each SRB also obtains two branches. Specifically, one branch is processed via a conventional residual layer, whereas the other undergoes a downsampled convolution first to enhance the richness of the feature representation. As shown in Fig. 2(a), each SRB contains multiple residual layers, with downsampling (via stride-2 convolution) and upsampling (via bilinear interpolation) performed only in the initial and final layers, respectively.

In addition, we incorporated the basic layer from Vision Mamba [34] in the decoder to refine the fused features F^i_f, which can further strengthen the correlation between complementary information and fill cloudy regions. Its bidirectional scanning mechanism effectively enhances the 2-D spatial-aware understanding. Furthermore, the advantages in local spatial information aggregation through Hilbert curve scanning have been demonstrated [43]. When dealing with sequence tokens, it can supplement the original bidirectional scanning order by promoting clustering of neighboring tokens. Therefore, the characteristic of long-range dependencies can assist the image in extracting information from a relatively long distance to fill large missing areas. Meanwhile, the utilization of local information improves the ability to suppress local outliers, such as the elimination of obvious cloudy boundaries.

Each LEM consists of consecutive processing by the two scanning mechanisms, named the Bi-Mamba layer and the Hi-Mamba layer. The specific steps of the Mamba layer are shown in Fig. 2(c). First, the 2-D features are flattened into 1-D sequences. By only adjusting the index of the 1-D embedding before the SSM and manually defining the scanning order, the establishment of the sequence-space correlation is adjusted. Before processing the sequence, normalize the input sequence through a layer normalization. This helps stabilize the training process and improve the model's performance. Then, it is linearly projected into two different spaces, x and z, which are used for the subsequent bidirectional processing branch and gating branch, respectively. In the bidirectional processing branch, each direction first employs a 1-D convolution to capture local dependencies and then conducts the SSM calculation. The outputs of the forward and backward SSMs are multiplied by the gated output of the z-space, which is achieved by SiLU activation. Subsequently, the results from the two directions are added, following a linear channel dimension mapping. Finally, a merging operation is required to convert 1-D sequences back to 2-D features. In this context, the steps of SSM are as follows.

The input sequence is dented as x, and the result of the 1-D convolution is x', which serves as the input for the SSM calculation. First transform x' by three linear layers to obtain three key parameters of the SSM: B, C, and Δ. $\overline{A}$ and $\overline{B}$ are derived from Δ. After preparing the parameters above, the final output y can be calculated through the SSM recurrent: $h_t = \overline{A}h_{t-1} + \overline{B}x_t$ and $y_t = Ch_t$, where t represents the time step that controls state transition.

C. Supervised Alignment Perception Strategy

The necessity of fine registration before fusion has been verified in [10]. However, existing methods have limitations such as instability and inaccuracy in deviation elimination. Therefore, we designed a SAPS. By adding manual deformable offsets to SAR features during the training phase, it imposes direct constraints on APM to guide it to focus on the critical deviations between the two modalities. Moreover, to further enhance the stability, we combined two typically used strategies [44]: the flow-guided and deformable alignment approaches. The flow-guided alignment approach models the deviation coordinates for each pixel and performs individual sampling adjustments. The deformable alignment approach, by adding offsets to the sampling positions of traditional convolution kernels to obtain variable receptive fields, captures adaptive contextual information to detect spatial misalignments. Specifically, the flow-guided approach

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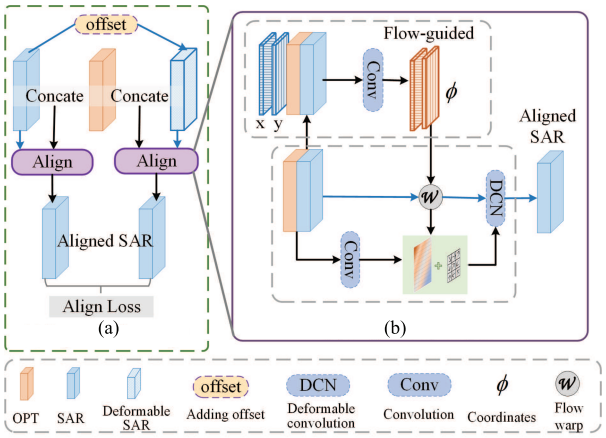


Fig. 3. (a) Details of SAPS, in which the polynomial bias is exclusively applied to SAR with the aim of introducing manual deviations in the training phase. (b) Specific construction of the APM.

provides a fundamental motion estimation for deformable alignment, enabling the DConv to adjust sampling positions not in a groundless manner, but by refining and optimizing locally based on the overall motion trends provided by optical flow [45].

First of all, as shown in Fig. 3(a), for the implementation of the supervised strategy, the concatenated F_{opt} and F_{sar} serves as the reference group G_{ref}. The deviation group G_{dev} is implemented with F_{opt} and features with manual offset deviations $F_{\text{sar}}^{\text{err}}$. Considering that high-order polynomials are capable of achieving more complex nonlinear transformations, and of enabling the modeling of more intricate geometric relationships between SAR and optical images, we adopted them to add the deformable offset. In 2-D space, the polynomial transformation can transform the original coordinates (u, v) into new coordinates (x, y). The formula is as follows:

$$\begin{cases} x = F_x(u,v) = \sum_{i=0}^{n} \sum_{j=0}^{n-i} a_{ij} u^i v^j \\ y = F_y(u,v) = \sum_{i=0}^{n} \sum_{j=0}^{n-i} b_{ij} u^i v^j \end{cases} \tag{1}$$

where n represents the number of terms in the polynomial, and a_{ij}, b_{ij} are the coefficients of each term. In application, the transformation parameters are obtained through random sampling, and the randomly deviated pixels are restricted within the range of [0,10]. Then, based on the aforementioned formulas, deformation coordinates for each pixel of F_{sar} can be calculated, and $F_{\text{sar}}^{\text{err}}$ are generated by position transformation and resampling.

Subsequently, G_{ref} and G_{dev} are, respectively, input into the APM to achieve F_{ref} and F_{dev}. Since the former represents the residual offset between modalities and the latter is the artificially introduced offset, we can apply L2 regularization to calculate align loss $\mathcal{L}_{\text{Align}}$

$$\mathcal{L}_{\text{Align}} = \sum_{i=0}^{2} \|F_{\text{ref}}^i - F_{\text{dev}}^i\|_2 \tag{2}$$

where i represents each scale corresponding to the encoder stage. This loss is used to constrain the training process of the APM, aiming to enhance the sensitivity to spatial deviations without over-eliminating key information.

The specific construction of the APM is shown in Fig. 3(b), which combines two aligning methods. For flow-guided alignment approach, the input $G_k, k \in \{\text{ref}, \text{dev}\}$ will be used to generate deviation coordinates ϕ with $(H, W, 2)$, where H and W are consistent with input size, and the two are, respectively, the obtained deviation amounts in the X-direction and the Y-direction. Next, G_{dev} will be taken as an example to introduce the APM. First, in the flow warp, accurate feature values for each pixel need to be located based on the predicted coordinate relationships. The warp operation is expressed as

$$F_{warp}^{\text{err}}(p) = F_{\text{sar}}^{\text{err}}(p + \phi_{\text{dev}}(p)) \tag{3}$$

where p represents each pixel location, and ϕ_{dev} denotes the predicted deviation coordinates from G_{dev}.

As the acquisition of ϕ emphasizes the sensitivity to horizontal and vertical directions, respectively, horizontal and vertical coordinate parameters initialized a priori are added compared with vanilla convolution [46]. These two learnable parameters are concatenated with the input features and then used to predict ϕ through convolution

$$\phi_{\text{dev}} = \text{Conv}_{(c+2)->2}\left(\left[G_{\text{dev}}, F_x, F_y\right]\right) \tag{4}$$

where F_x and F_y denote the initial single coordinate parameters; $\text{Conv}_{(c+2)->2}$ represents the channel dimension transformation in the convolution; [,] is concatenation.

Correspondingly, on the other side of the flow-guided branch, we referred to the structure of DConv [11], which consists of two steps: convolution kernel's offset acquisition and offset-guided DConv. The offset with a size of $(H, W, 2k^2)$ is predicted based on G_{dev}, where k is convolution kernel size. The ϕ is also used to constrain this offset parameter, which is then viewed as the receptive field deformation for DConv on F_{warp} to obtain the final aligned F_{dev}. This process can be formulated as

$$F_{\text{dev}}(p) = \sum_{pk} w_k \cdot F_{warp}(p + \Delta pk + \phi(p)) \cdot \Delta m_k \tag{5}$$

where w_k is DConv weight; p_k is the relative position in the kernel, and Δm_k is a modulation scalar for location p corresponding to p_k.

As shown in Algorithm 1, we solely relied on the APM to perceive and rectify the deviations between the two inputs. Moreover, the above operations are carried out for each encoder stage to dynamically adjust the feature representations of SAR at different scales.

D. Modality Difference Selection Module

In fusion design, it is necessary to highlight the mutual guidance ability of the two modalities to improve the utilization of complementary information. Therefore, an adaptive fusion module based on a difference mechanism is introduced to dynamically regulate SAR information integration. Specifically, in cloud-dense areas, the model boosts the confidence of SAR data and adjusts fusion weights. Conversely, the weights of optical features will be improved.

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Algorithm 1 1 SAPS

Input: SAR feature: F_{sar}, Optical feature: F_{oppt};
Output: Rectified SAR feature F_{ref}.

if training **then**
/*Manual Offset*/
$F_{sar}^{err} = \text{Poly}(F_{sar})$
/*Reference and deviation groups*/
$G_{ref} = [F_{opt}, F_{sar}], G_{dev} = [F_{opt}, F_{sar}^{err}]$
for *i in [ref, dev]* **do**
Compute F_i *from Eq.(3) to Eq.(5)*
end for
Compute L_{align} *by Eq.(2)*
else
Compute F_{ref} *from Eq.(3) to Eq.(5)*
end if
return F_{ref}

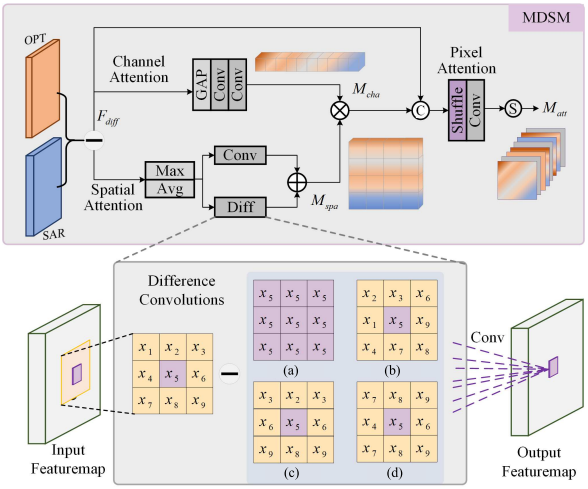


Fig. 4. Details of MDSM. (a) Central difference convolution. (b) Angular difference convolution. (c) Vertical difference convolution. (d) Horizontal difference convolution.

First, a differential calculation between F_{opt} and $F_{\text{sar}}^{\text{ref}}$ is performed to enhance the discriminative information. Then, the differences F_{diff} are measured from the spatial, channel, and pixel perspectives to promote weight measurement between F_{opt} and $F_{\text{sar}}^{\text{ref}}$. Considering the geometric consistency of ground object between the two modalities, we focused on importance modeling from a spatial perspective. The key edge features are highlighted through differential convolution [47], which has a remarkable representational ability for fine-grained features, and thus is suitable for enhancing structural features in spatial dimension. As shown in Fig. 4, the initial F_{diff} pass through parallel branches to obtain the channel attention map M_{cha} and the spatial attention map M_{spa}, which interact with each other multiplicatively. The M_{cha} is measured by global average pooling and two linear projections. For M_{spa}, we measured the spatial importance from four different aspects: central [Fig. 4(a)], angular [Fig. 4(b)], horizontal [Fig. 4(c)], and vertical [Fig. 4(d)] difference. Furthermore, through weight parameter aggregation [48], the aforementioned advantages of four differential convolutions can be leveraged under the same computational complexity as vanilla convolution

$$w_i = \sum_{j=1}^{5} w_{ij}, b_i = \sum_{j=1}^{5} b_{ij} \tag{6}$$

where w_i and b_i represent the aggregated convolutional kernel weight and bias of each position i, and j represents the kernels of vanilla convolution and four differential convolutions. Then different convolution is operated based on the aggregated w_i.

Subsequently, pixel attention mechanism strengthens the region importance measurement. Compared with directly concatenating along the channel dimension, it expands the dimension of the input features first to avoid the information aliasing of each channel. After this channel shuffle, a group convolution operation is employed to ensure channel-wise independence and more refined information.

With the above processes, the final attention map M_{att} can be formulated as

$$M_{\text{att}} = \sigma\left(G\text{Conv}_{7\times7}\left(\text{shuffle}\left(F_{\text{diff}}, M_{\text{cha}} \otimes M_{\text{spa}}\right)\right)\right) \tag{7}$$

where σ is sigmoid function, $G\text{Conv}_{7\times7}$ represents the grouped convolution with a kernel size of 7, shuffle is dimensional expansion and concatenation, and $\otimes$ is matrix multiplication operation. Guided by $M_{\text{att}}, F_{\text{fus}}$ is obtained as follows:

$$F_{\text{fus}} = M_{\text{att}} \odot F_{\text{sar}}^{\text{rec}} + (1 - M_{\text{att}}) \odot F_{\text{opt}}. \tag{8}$$

E. Loss Function

The loss calculation consists of two parts: the reconstruction loss $\mathcal{L}_{\text{Rec}}$ between the reconstructed cloud-free image I_{rec} and the real cloud-free image I_{gt}, as well as the alignment loss $\mathcal{L}_{\text{Align}}$ introduced in Section III-C. The total loss function $\mathcal{L}_{\text{total}}$ for network training is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Rec}} + \lambda\mathcal{L}_{\text{Align}} \tag{9}$$

where λ is a hyperparameter used to balance between the two losses. The $\mathcal{L}_{\text{Rec}}$ includes the L1 loss $\mathcal{L}_{L1}$ and the structural similarity index measure (SSIM) loss $\mathcal{L}_{\text{ssim}}$. The $\mathcal{L}_{L1}$ can directly constrain the reconstructed pixel values and is capable of reconstructing clear detailed information, which is formulated as

$$\mathcal{L}_{L1} = \|I_{\text{gt}} - I_{\text{rec}}\|_1. \tag{10}$$

The $\mathcal{L}_{\text{ssim}}$ measures the similarity of images from three dimensions: brightness, contrast, and structure, and can effectively capture the structural information

$$\mathcal{L}_{\text{ssim}}\left(I_{\text{gt}}, I_r\right) = 1 - \text{SSIM}\left(I_{\text{gt}}, I_r\right). \tag{11}$$

IV. Experiments and Results

A. Experimental Settings

1) Datasets: We selected three datasets for experimental validation: SEN12MS-CR [49], LuojiaSET-OSFCR [40], and WHU-OPT-SAR [50]. The details are as follows.

SEN12MS-CR is the most widely used dataset, which was collected from the Sentinel-1/2 satellite platform. With its extensive scene coverage and rich cloud states, this dataset effectively validates the generalization capabilities across diverse scenarios. The paired triplet data are constrained in the

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TABLE I
QUANTITATIVE COMPARISONS ON THREE DATASETS

Method	SEN12MS-CR				LuojiaSET-OSFCR				WHU-OPT-SAR			
	PSNR↑	SSIM↑	SAM↓	MAE↓	PSNR↑	SSIM↑	SAM↓	MAE↓	PSNR↑	SSIM↑	SAM↓	MAE↓
DSen2-CR	28.2882	0.8855	4.5254	0.0324	35.2004	0.9363	2.3386	0.0133	32.1217	0.9316	1.2235	0.0206
GLF-CR	28.5467	0.8853	4.8228	0.0319	35.0866	0.9305	2.4235	0.0133	34.0562	0.9270	1.1105	0.0170
USSRN-CR	28.7649	0.8930	4.5580	0.0320	35.7301	0.9463	2.1953	0.0127	34.2447	0.9408	1.0706	0.0172
HS2P	29.0529	0.9001	4.4201	0.0304	35.7524	0.9409	2.1620	0.0126	34.6118	0.9387	0.9763	0.0163
Align-CR	28.6625	0.8868	4.9089	0.0324	35.1635	0.9305	2.3727	0.0132	34.7700	0.9328	1.0024	0.0157
DDIA-CFR	29.1671	0.9060	4.2527	0.0316	35.0905	0.9386	2.4279	0.0135	34.7729	0.9386	1.0803	0.0168
HDRSA-Net	28.8115	0.8944	4.7250	0.0321	34.5040	0.9294	2.5094	0.0142	34.0758	0.9321	1.2312	0.0173
HPN-CR	28.9807	0.8983	4.8797	0.0312	34.9053	0.9338	2.6110	0.0142	34.6675	0.9363	1.1929	0.0164
Ours	30.0112	0.9124	4.1375	0.0277	36.2452	0.9475	2.0320	0.0118	35.4147	0.9440	0.9964	0.0157

same season and are all resampled to 10 m resolution. Based on the division of regions, random sampling was performed, resulting in 12 000 samples for the training set and 4000 samples each for the test and validation sets.

LuojiaSET-OSFCR is also collected from the Sentinel platform. In addition, it provides cloud masks for each cloudy image, which assist in validating cloud removal performance under varying cloud conditions. Furthermore, land cover annotations are included to verify semantic recovery performance across different land types. Comprising 20 000 image pairs, the dataset was randomly sampled at a 6:2:2 ratio into training, validation, and test sets.

Furthermore, to validate the generalization capabilities of cloud removal methods across different satellite, experiments were also conducted on WHU-OPT-SAR collected from the GaoFen-1/3 satellites. This dataset contains four-band optical imagery and single-polarization SAR imagery. In addition, the cloudy optical images were simulated via Perlin noise. The dataset was partitioned according to distinct regions of interest into training (17 640), test (5880), and validation (5880) sets.

2) Evaluation Metrics: We utilized the peak signal-to-noise ratio (PSNR), the SSIM, the spectral angle mapper (SAM), and the mean absolute error (MAE) to quantitatively compare different methods. For the cloud removal task, the reconstructed images are expected to achieve higher PSNR and SSIM while reducing SAM and MAE.

3) Implementation Details: Our experiments utilize PyTorch framework on an Ubuntu environment with an NVIDIA GeForce RTX 4090 GPU. We employed the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a learning rate of 1×10^{-4} (decayed by half every 5 epochs) to optimize the network training. The batch size is set to 6, and the maximum number of training rounds is set to 30 epochs.

B. Comparison Experiments

To validate the effectiveness of the proposed SADS-CR, eight representative state-of-the-art approaches were selected, including DSen2-CR [38], GLF-CR [12], USSRN-CR [14], HS2P [13], Align-CR [10], DDIA-CFR [51], HDRSA-Net [40], and HPN-CR [32]. Experimental validation was conducted from five aspects: quantitative metric evaluation, qualitative visual analysis, validation under varying cloud cover levels, reconstruction performance across different land cover types, and performance analysis under modal misalignment conditions. Detailed experimental results are presented below.

1) Quantitative Comparisons: The quantitative evaluation results are presented in Table I. SADS-CR achieves the best reconstruction performance and the least reconstruction MAE on all three datasets, with values of 0.0277, 0.0117, and 0.0157, respectively. The proposed method improves by 0.8, 0.01, and 0.12 in PSNR, SSIM, and SAM metrics, respectively, compared to the suboptimal DDIA-CFR on the SEN12MS-CR dataset. For LuojiaSET-OSFCR, our method improves the PSNR score from 35.7524 to 36.2554 and enhances the SSIM score from 0.9409 to 0.9475, compared to HS2P. Although the proposed method demonstrates a significant leading advantage in PSNR for reconstruction performance on the WHU-OPT-SAR dataset, its performance in the SAM metric is slightly inferior to HS2P. This might be primarily because HS2P's joint structure-spectral preservation strategy effectively enhances spectral information recovery. However, our method prioritizes structural recovery.

2) Qualitative Comparisons: We selected four pairs of reconstructed images of different scenes from each dataset for qualitative analysis. As shown in Fig. 5, four examples from SEN12MS-CR, respectively, illustrate the interference of clouds in different states. The clouds in the first row exhibit a small and fragmented distribution, with both thin and thick clouds coexisting. This distribution pattern easily leads to uneven reconstruction, overcorrection in thin cloud areas, and insufficient filling in thick cloud areas. For example, the reconstruction results of columns in Fig. 5(a)–(c) clearly show excessive brightness in the thin cloud areas. In contrast, our results demonstrate a more uniform overall reconstruction. The lower half of the images in the second row exhibit large-scale thick cloud accumulation. In such scenarios, the textures of clouds often lead to unrealistic artifacts in the reconstructed imagery [as shown in columns in Fig. 5(c), (e), and (g)]. For Align-CR, such complex textures of the background terrain further exacerbate the instability of the align module. With the assistance of the supervised alignment strategy and fusion module, SADS-CR can effectively utilize the real texture information in SAR while achieving suppression of thick cloud textures. The third-row samples feature block clouds that disrupt the intact boundaries of cultivated land. Most networks result in significant discrepancies between the reconstructed region and its surroundings. This is primarily attributed to their

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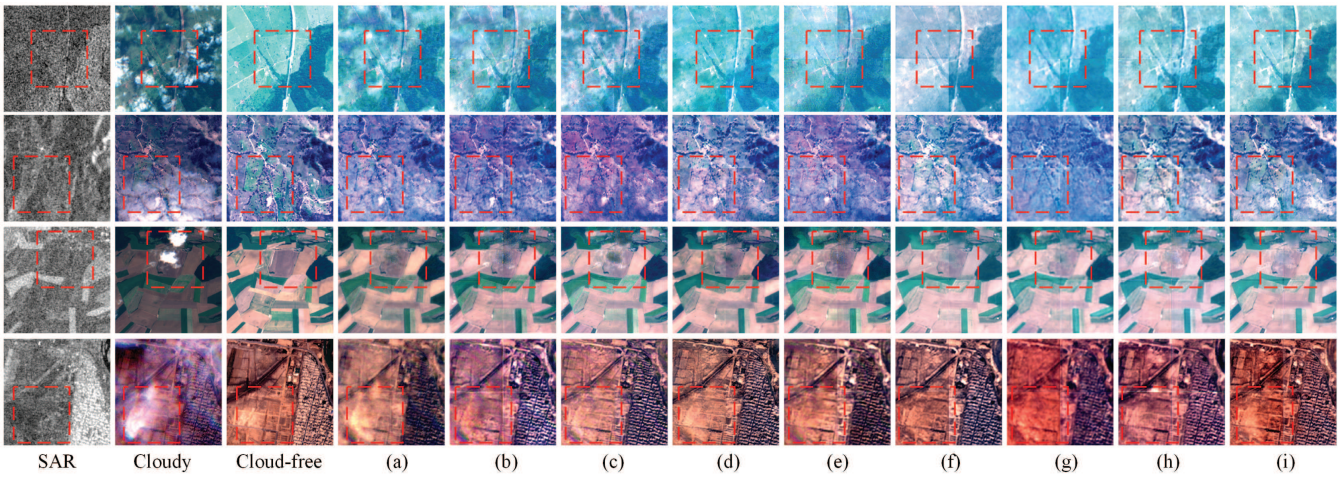


Fig. 5. Qualitative comparison on SEN12MS-CR. The red boxes emphasize the areas of clear quality contrast in the reconstruction results. (a)–(i) Reconstruction results of DSen2-CR, GLF-CR, USSRN-CR, HS2P, Align-CR, DDIA-CFR, HDRSA-Net, HPN-CR, and ours.

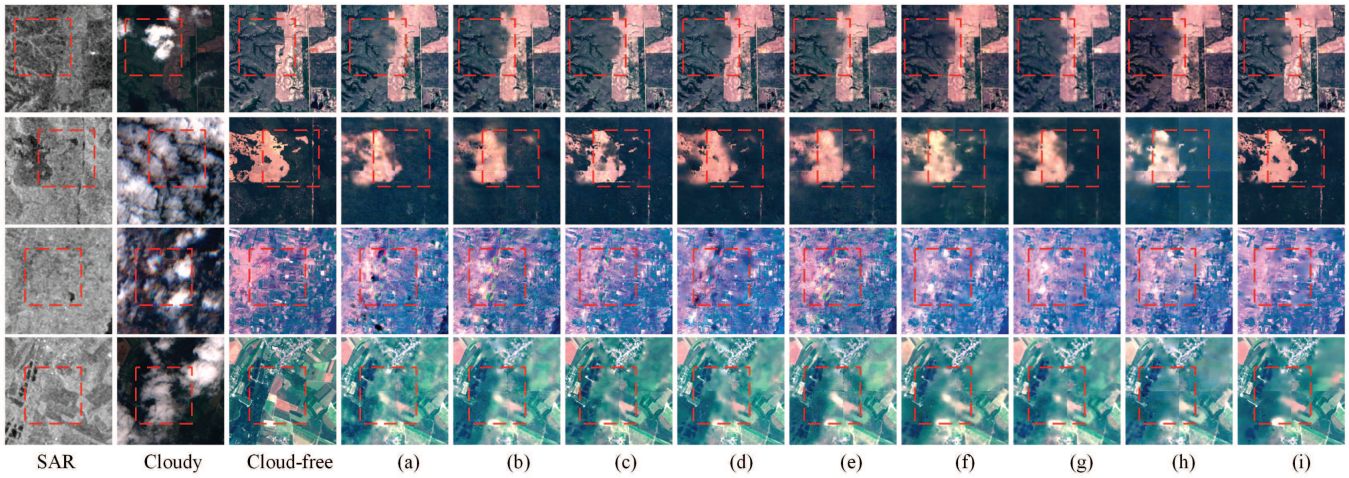


Fig. 6. Qualitative comparison on LuojiaSET-OSFCR. The red boxes emphasize the areas of clear quality contrast in the reconstruction results. (a)–(i) Reconstruction results of DSen2-CR, GLF-CR, USSRN-CR, HS2P, Align-CR, DDIA-CFR, HDRSA-Net, HPN-CR, and ours.

failure to adequately identify the damaged areas, leading to deformable reconstruction boundaries under the interference of cloudy pixels. Furthermore, thick cloud reconstruction requires refinement, which is achieved through a combination of SAR information assistance and improved understanding of the cloudy area.

The visualization of the Luojia-OSCFR dataset is shown in Fig. 6. It is obvious that our reconstruction results exhibit clearer edges and preserve fine-grained texture structures. Similarly, as shown in the column of Fig. 6(c), USSRN-CR also achieves clear structure restoration due to its adoption of edge maps of SAR. Although the overlap of cloud boundaries and ground object boundaries significantly increases the difficulty of reconstruction, the proposed MDSM can restrain this boundary distortion. Furthermore, as shown in the third row, our reconstruction results exhibit stronger suppression capability for outlier pixels that manifest as colorful stripes. In contrast, the simple attention filter in HS2P fails to mitigate the impact of outlier pixels. Furthermore, the frequency domain features introduced by DDIA-CFR, the sparse global context aggregation module in HDRSA-Net, and the combination of CNN and transformer in HPN-CR can all integrate global information very well, thus better preserving the overall integrity. Nonetheless, their performance remains limited in scenarios with thick clouds. In contrast, our SADS-CR employs an LEM layer, achieving the refinement of complete boundary structures and central area filling under thick cloud conditions.

As illustrated in Fig. 7, the images in WHU-OPT-SAR distinguish ground objects more clearly compared to the above two datasets. Although the simulated clouds fail to entirely reflect the diffuse boundaries and irregular textures of real clouds, they do damage the quality of optical images and can be used for cloud-removal verification. Overall, the differences between the comparison methods are not significant, and our advantage is only evident in certain detailed reconstructions. For example, the reconstruction results of thick cloud regions in the third row exhibit uneven filling, with the results in the columns of Fig. 6(c), (e), and (h) being the most severe. In contrast, while HS2P and our reconstruction demonstrate

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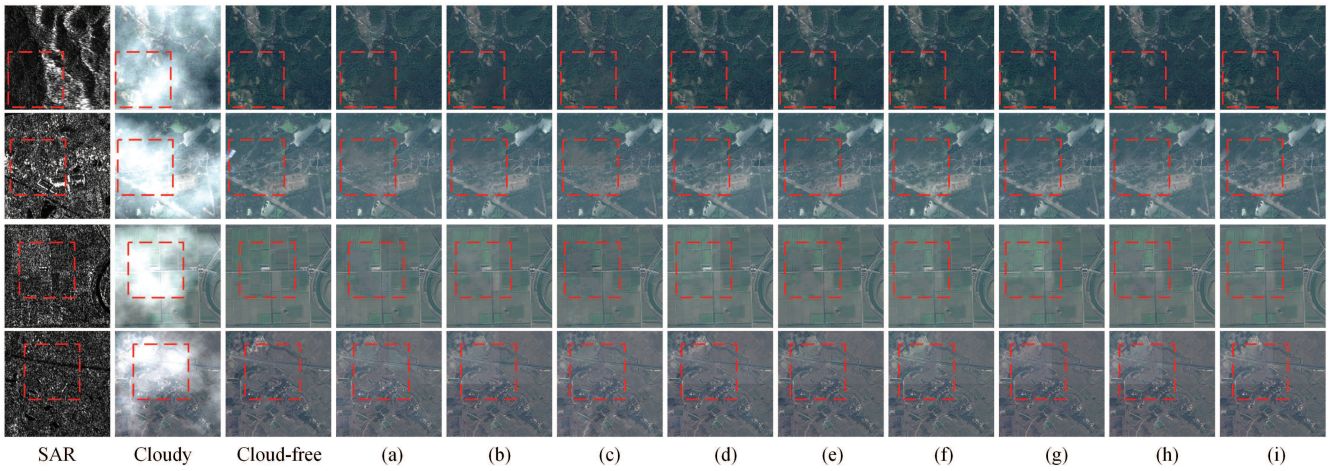


Fig. 7. Qualitative comparison on WHU-OPT-SAR. The red boxes emphasize the areas of clear quality contrast in the reconstruction results. (a)–(i) Reconstruction results of DSen2-CR, GLF-CR, USSRN-CR, HS2P, Align-CR, DDIA-CFR, HDRSA-Net, HPN-CR, and ours.

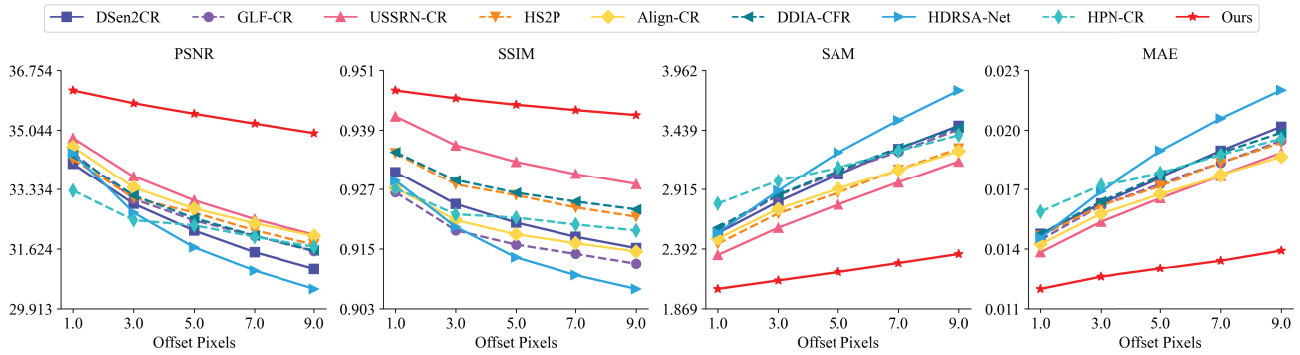


Fig. 8. Metrics performance on the LuojiaSET-OSFCR dataset under different misaligned conditions.

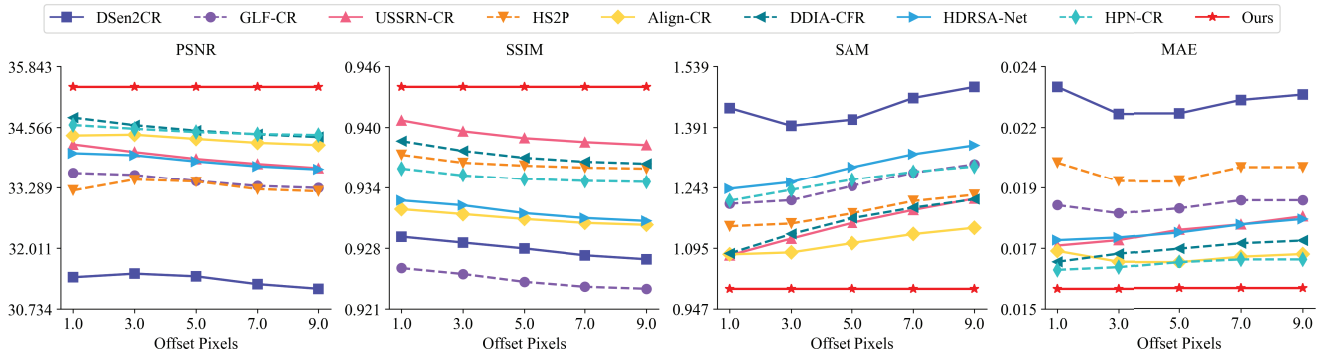


Fig. 9. Metrics performance on the WHU-OPT-SAR dataset under different misaligned conditions.

slightly spectral error, they are capable of identifying the boundaries of farmland obscured by clouds. Furthermore, most methods are unable to leverage structural information from SAR, thereby causing the fine details in cloud-region reconstructions to appear excessively smooth. By comparison, our method not only maintains global structural consistency but also preserves fine-grained local details within densely clouded central regions.

3) Analysis of Spatial Misalignment: To further validate the performance under misalignment conditions, we introduced polynomial deformations to simulate image-level spatial misalignment. These deformations contained complex deformable transformations to meet real-world scenarios. As shown in Fig. 8, the proposed method consistently maintains the best performance under the bias conditions ranging from 1 to 9 pixels, without experiencing a substantial performance decline. Specifically, the fluctuation ranges of the four metrics are 1.2, 0.005, 0.3, and 0.002, respectively. This demonstrates the necessity of bias elimination before fusion, and the offset perception strategy we designed can alleviate the residual misalignment. Although there is a performance degradation under offset conditions, our method still has a significant

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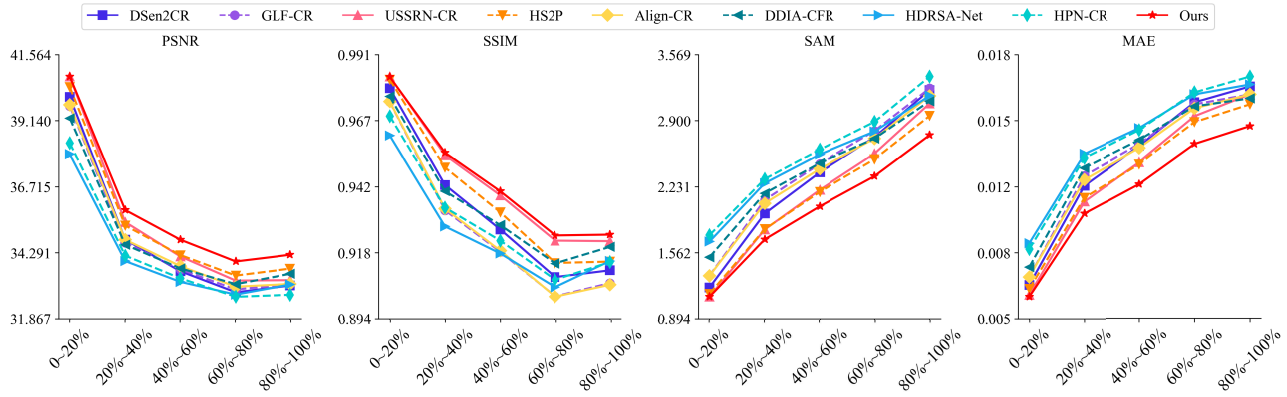


Fig. 10. Line charts of cloud removal metric performance under different cloud cover levels.

TABLE II
SEGMENTATION VALIDATION OF RECONSTRUCTED OPTICAL IMAGES ON LUOJIASET-OSFCR AND WHU-OPT-SAR

Method	LuojiaSET-OSFCR				WHU-OPT-SAR			
	Acc ↑	F1 ↑	Kappa ↑	MIoU ↑	Acc ↑	F1 ↑	Kappa ↑	MIoU ↑
cloudy	0.406	0.376	0.296	0.243	0.700	0.438	0.541	0.305
cloud-free	0.843	0.814	0.804	0.679	0.815	0.560	0.717	0.434
DSen2-CR	0.762	0.728	0.702	0.592	0.793	0.520	0.684	0.389
GLF-CR	0.767	0.734	0.708	0.599	0.795	0.521	0.687	0.391
USSRN-CR	0.762	0.725	0.701	0.589	0.791	0.522	0.681	0.391
HS2P	0.771	0.736	0.713	0.601	0.787	0.524	0.673	0.393
Align-CR	0.772	0.739	0.715	0.604	0.795	0.521	0.687	0.391
DDIA-CFR	0.753	0.717	0.690	0.581	0.793	0.523	0.685	0.392
HDRSA-Net	0.758	0.719	0.697	0.581	0.794	0.523	0.686	0.392
HPN-CR	0.740	0.696	0.673	0.560	0.795	0.524	0.687	0.394
Ours	0.782	0.747	0.727	0.614	0.797	0.526	0.690	0.396

advantage in terms of antibias capability. HPN-CR also displays a slight advantage in antibias capability. This may be attributed to the adoption of the transformer architecture in its feature extraction process. The utilization of global information can suppress the influence of spatial biases. In contrast, the performance degradation of HDRSA-Net is the most obvious. This is likely because when the spatial domain features with biases are transformed into the frequency domain, the frequency domain features amplify these biases. The incorrect frequency domain information fails to accurately represent the target features in the image. However, manual offset scenarios reveal that Align-CR's antibias performance is not notably superior. This deficiency likely stems from the instability of the alignment strategy based on DConv, which fails to accurately model intermodality spatial bias.

Similarly, the results under misalignments on the WHU-OPT-SAR with higher spatial resolution are shown in Fig. 9. Although high-resolution images have higher requirements for registration accuracy, the information content represented by a single pixel decreases. Therefore, the vast majority of methods can suppress the offset based on contextual information and do not exhibit significant performance degradation. Furthermore, the proposed method still maintains a significant advantage, generally showing a horizontal trend in all offset conditions. This indicates that the SAPS exhibits strong stability and generalization ability to handle images of different spatial resolutions.

4) Analysis of Cloud Cover Levels: The further analysis of different cloud coverages is based on the existing five types of cloud amount stratifications in LuojiaSET-OSFCR. The line chart is depicted in Fig. 10; as the cloud amount increases, the reconstruction indicators exhibit a downward trend. Under the level of 0%–20% cloud cover, the difference in reconstruction performance between our method and others is relatively tiny. This phenomenon stems from the limited contribution of the fusion design under sparse cloud coverage, where the reconstructed image predominantly relies on optical data and thus insufficiently leverages the benefits of cross-modality integration. As the cloud cover level increases, this gap widens, which shows that the perception and fusion strategy in our SADS-CR has a stronger ability to compensate for the situation where a large amount of information in the optical image is missing. In conclusion, the proposed method is capable of dealing with the interference of information loss in various situations, highlighting the generalization ability of the model.

5) Analysis of Different Land Cover Types: To validate the semantic recovery capability of the reconstruction results, we used the pretrained DeepLabv3+ model (trained on cloud-free optical imagery) to segment the reconstructed cloud-free imagery in LuojiaSET-OSFCR. We selected accuracy, F1-score, kappa coefficient, and MIoU as segmentation metrics, which are shown in Table II. SADS-CR obtains the best segmentation results, while Align-CR achieved suboptimal

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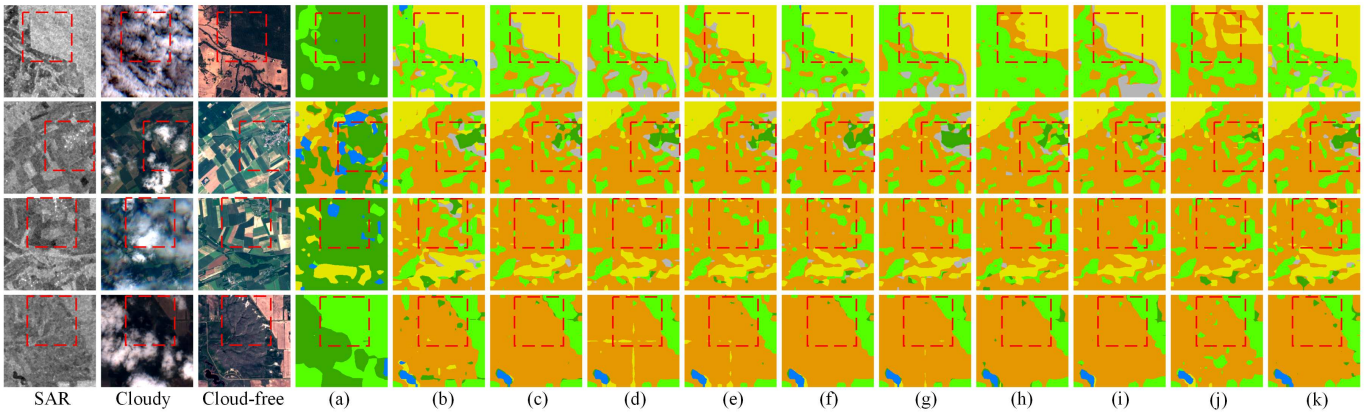


Fig. 11. Comparison of semantic recovery on LuojiaSET-OSFCR dataset. (a) Segmentation results of cloudy images. (b) Segmentation results of cloud-free images. (c)–(k) Segmentation results of images reconstructed through DSen2-CR, GLF-CR, USSRN-CR, HS2P, Align-CR, DDIA-CFR, HDRSA-Net, HPN-CR, and ours.

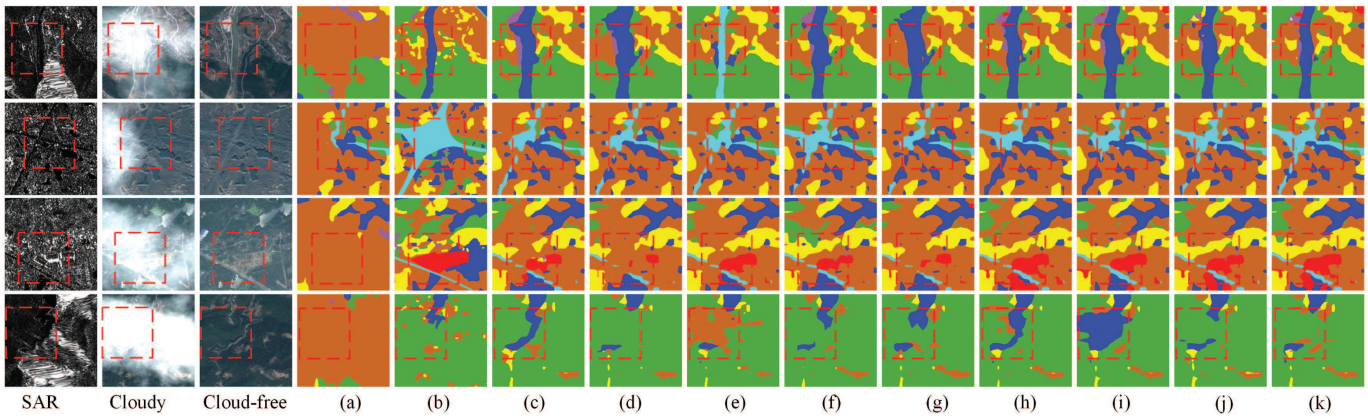


Fig. 12. Comparison of semantic recovery on WHU-OPT-SAR dataset. (a) Segmentation results of cloudy images. (b) Segmentation results of cloud-free images. (c)–(k) Segmentation results of images reconstructed through DSen2-CR, GLF-CR, USSRN-CR, HS2P, Align-CR, DDIA-CFR, HDRSA-Net, HPN-CR, and ours.

segmentation performance. This indicates that subsequent land cover classification applications require even higher spatial accuracy from reconstruction results. Furthermore, the segmentation accuracy of all reconstruction results lies between that of cloudy and cloud-free imagery, indicating that cloud removal can recover underlying land feature information to some extent, but this recovery still has limitations. Furthermore, the performance on the WHU-OPT-SAR dataset discrepancies among the comparative methods are marginal, with the segmentation accuracy uniformly around 0.790. However, achieving the best performance in all metrics demonstrates the effectiveness of our method. In contrast, despite achieving top performance in the SAM metric, HS2P underperforms in segmentation tasks. The reason for this could be that four-band optical data segmentation emphasizes spatial geometric features over spectral correlations, which play a relatively subordinate role.

In addition, qualitative segmentation results are shown in Figs. 11 and 12. From the segmentation results of cloudy imagery, the clouds not only reduce the clarity of land feature boundaries but also induce large-scale land feature confusion. Cloud removal methods effectively tackle such challenges, and the segmentation accuracy has been significantly improved. Especially, the proposed method can minimize land feature confusion caused by large-scale thick clouds and reduce misclassification and omission of land features at cloud margins, making the segmentation results closer to those of cloud-free imagery. As shown in the first row of WHU-OPT-SAR dataset, clouds completely obscure the boundary between forest and farmland and erase water bodies. By contrast, the reconstruction results not only eliminate such interference but also fully reconstructs the water body with precise boundaries. Furthermore, the cloudy areas are highly prone to being classified as farmland, likely because cloudy pixels have high pixel values and appear in large areas, which are similar to farmland distribution features. Through above experiments, we verified the generalization ability of the proposed method under different terrains and the feasibility of cloud removal for downstream applications.

6) Analysis of Parameters and Time Cost: To assess the practical efficiency and feasibility of the proposed method, we compared the network parameter counts and inference time on the test set of LuojiaSET-OSFCR using an NVIDIA RTX 4090. The experimental results are shown in Fig. 13. Overall, our method demonstrates moderate performance in both parameter count and computational efficiency.

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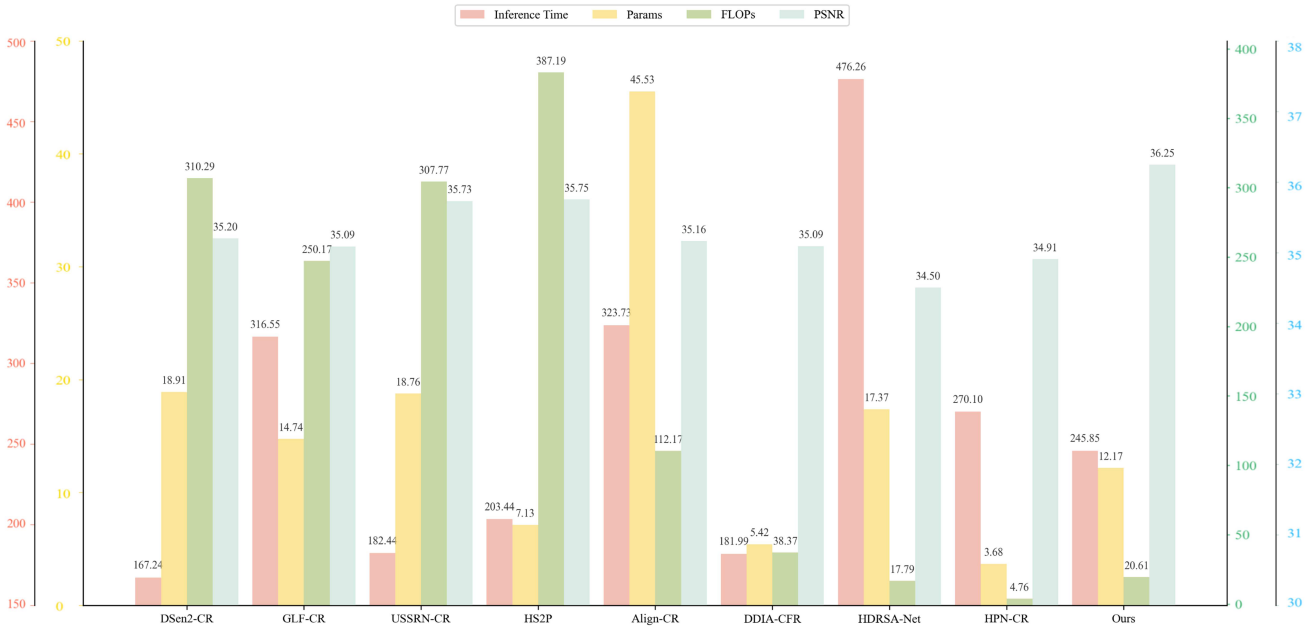


Fig. 13. Comparison among the running time, computational amount, and number of parameters of different models.

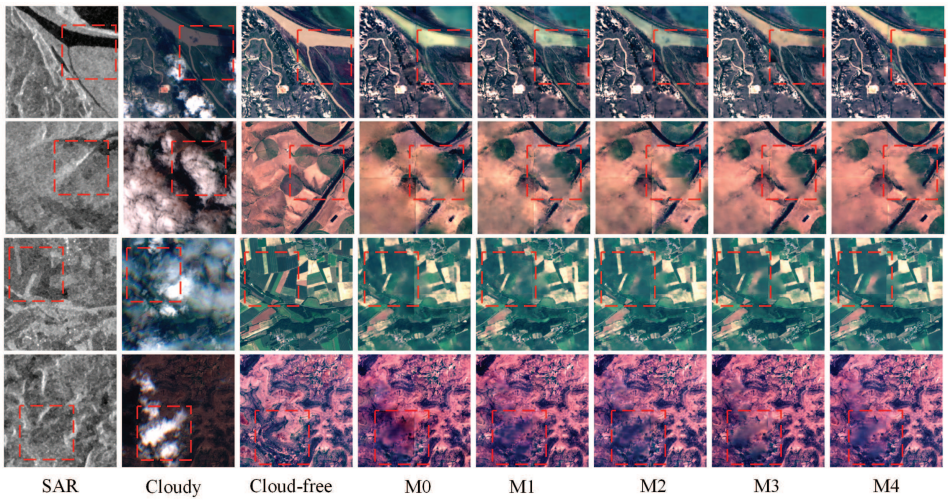


Fig. 14. Qualitative results of ablation experiments.

The relatively high time overhead stems from the necessary alignment perception prior to fusion. Nonetheless, considering the performance gains, this trade-off is reasonable. Notably, while Align-CR accounts for the impact of spatial misalignment on reconstructed image quality, its multiscale alignment design leads to notable increases in both model parameters.

C. Ablation Experiments

We performed ablation experiments on the LuojiaSET-OSFCR dataset by systematically adding individual components of the additional modules to the basic setting. The results of the five ablation model settings are shown in Table III. The basic structure, composed exclusively of SRB, serves as M0. Subsequently, we successively obtained M1–M3 through the following replacements: replacing the decoder with LEM, substituting the MDSM with simple concatenation of multimodal features, and replacing the simple concatenation with SAPS. M4 represents our whole network, which improved the PSNR and SAM by 0.32 and 0.14, respectively, compared to the basic setting, demonstrating the effectiveness of the proposed modules. In addition, four representative samples are provided in Fig. 14, showcasing different scenarios such as farmland,

TABLE III
QUANTITATIVE ABLATION RESULTS ON LUOJIASET-OSFCR DATASET

Settings	Param(M)	PSNR↑	SSIM↑	SAM↓	MAE↓
M0	12.369	35.7086	0.9441	2.1774	0.0125
M1	**11.862**	35.8340	0.9457	2.1476	0.0124
M2	12.015	35.9749	0.9463	2.1072	0.0121
M3	12.104	35.9328	0.9463	2.1003	0.0122
M4 (Ours)	12.171	**36.2452**	**0.9475**	**2.0320**	**0.0118**

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TABLE IV
ABLATION RESULTS ON THE NUMBER OF SRB

Numbers	Param(M)	PSNR↑	SSIM↑	SAM↓	MAE↓
2	9.071M	35.7612	0.9451	2.1491	0.0125
4 (Ours)	12.171M	**36.2452**	**0.9475**	**2.0320**	**0.0118**
6	15.271M	35.6732	0.9454	2.1744	0.0126

TABLE V
ABLATION RESULTS ON THE NUMBER OF DOWNSAMPLING STAGES

Stages	Param(M)	PSNR↑	SSIM↑	SAM↓	MAE↓
1	1.878	35.0553	0.9428	2.3541	0.0137
2	5.330	35.4843	0.9453	2.2083	0.0129
3 (Ours)	12.171	**36.2452**	**0.9475**	**2.0320**	**0.0118**
4	28.512	35.8210	0.9455	2.1536	0.0124

city, mountain, and river. As illustrated in the last row, there is a significant difference between the cloudy area and the surrounding environment in M0's reconstruction. In contrast, the LEM introduced in M1 alleviates spectral discrepancies stemming from the inherently limited global information capture capability of traditional CNN-based structures. As shown in the third row, the reconstructed farmland of M4 appears most clearly, primarily due to the integration of MDSM and SAPS. The advantages can be attributed to mitigating interference from dense cloud textures and incorporating complete structural information from SAR. Furthermore, the alignment perception mechanism rectifies the deformation and residual registration errors in SAR imagery.

1) Analysis of Basic Structural: To comprehensively analyze the basic structural design in the network architecture, we conducted ablation experiments to evaluate the impact of residual layer counts in SRB and the number of downsampling stages. The outcomes concerning the SRB design are illustrated in Table IV, which indicate that the optimal number of basic SRBs is 4. This setting not only achieves the best performance but also exhibits a minor advantage in parameter counts.

We investigated the number of downsampling stages in the encoder to explore the rationality of its design, and the results are displayed in Table V. Given that each downsampling operation doubles the channel dimensions, a four-stage downsampling design leads to a substantial increase in parameters without commensurate gains in reconstruction performance. With three downsampling stages, the network has 12.171 M parameters and achieves a PSNR increase of approximately 0.76 compared to the two-stage design. Therefore, we balanced reconstruction accuracy and the number of network parameters, finalizing the number of downsampling stages as 3.

2) Analysis of the Selective Mechanism in Fusion: To further validate the adaptive adjustment capability, we visualized the feature maps generated by MDSM. As shown in Fig. 15, SAR and optical images reflect the consistent structural information of ground objects, and attention strength is observed to vary dynamically with different cloud states. The first row shows the thick cloud completely masking farmland boundaries. With MDSM-assisted fusion, a clearer

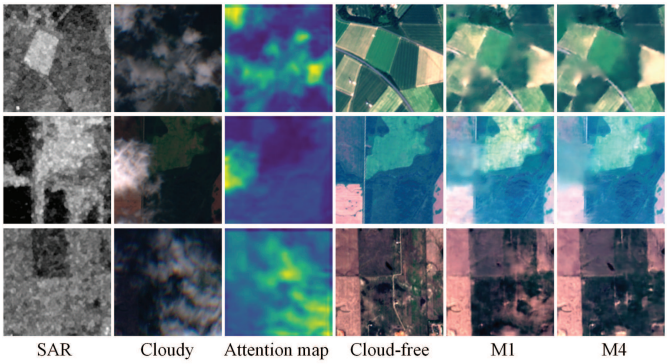


Fig. 15. Feature map visualization and comparison of the reconstruction results between M1 and M4.

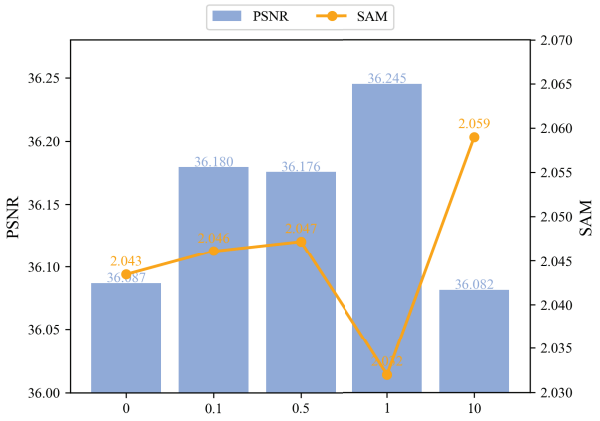


Fig. 16. Comparison and selection of loss ratio.

and more complete boundary reconstruction can be obtained, which benefits from the inhibition of the attention map on the cloud boundary. The comparison of the reconstruction results between M1 and M4 illustrates the ability of the attention map to select features and suppress noise for the two modalities during the fusion process.

3) Analysis of SAPS: To demonstrate the effectiveness of offset elimination, we conduct an analysis of the balance between $\mathcal{L}_{\text{Align}}$ and $\mathcal{L}_{\text{Rec}}$, as well as the manual offset settings during the training process. The hyperparameter analysis concerning the ratio between two losses is shown in Fig. 16, which includes five groups of different parameters. Among them, setting the ratio to 0 implies that artificial offsets do not participate in network optimization. The poor performance in this case validates the necessity of both the supervision strategy and artificial offsets. However, when the ratio is set to 10, the network training pays too much attention to the elimination of the offset; excessive emphasis on offset elimination during network training hinders the primary task, leading to a significant drop in both PSNR and SAM values.

Since there is inherent residual misalignment between optical and SAR images, the manually added offsets are crucial for the training stability and the spatial bias sensitivity. The manually configured misalignment is controlled by the offset parameter passed into the polynomial transformation model,

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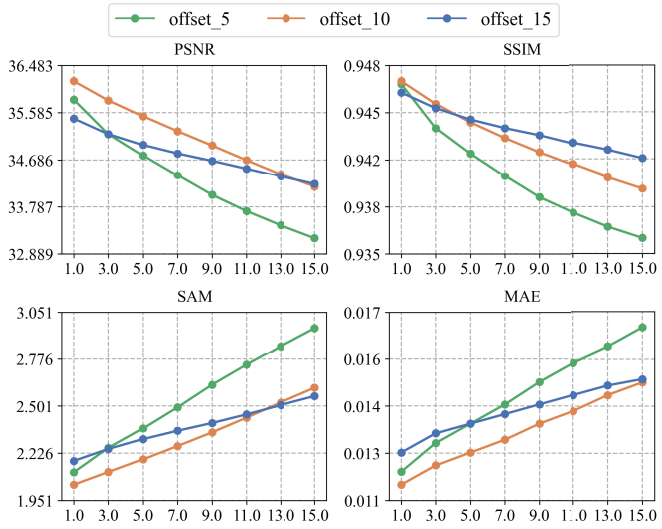


Fig. 17. Comparison of manual offset settings in training phase.

TABLE VI
MANUAL OFFSET SETTING

Offset (pixel)	PSNR↑	SSIM↑	SAM↓	MAE↓
5	36.1444	0.9463	2.0573	0.0119
10	36.2452	0.9475	2.0320	0.0118
15	35.6303	0.9468	2.1695	0.0127

and the comparison of the offset parameter is shown in Table VI. We analyzed the construction performance of the network under biases of 5, 10, and 15 pixels. Furthermore, Fig. 17 further illustrates the changes in the indicators under different offset conditions. While the 15-pixel setting achieves more stable and broader bias resistance capabilities, it affects the perception capability of the initial residual offsets, degrading reconstruction performance. Although the offset setting is strongly related to the antibias range, our method targets residual deviation persisting after initial registration. Considering both reconstruction performance and bias suppression effect comprehensively, we determined the manual offset to 10 pixels.

V. CONCLUSION

This article proposes a cloud removal network that integrates spatial alignment perception and modality difference selection in SAR-optical fusion, as well as global-local optimization in reconstruction. To mitigate spatial discrepancies between modalities, the SAPS contains two alignment perception mechanisms. Moreover, it enhances the training stability and sensitivity to initial biases through the addition of manual polynomial offsets. To filter out irrelevant information from modality-specific noise, the MDSM focuses on the differences in spatial dimension between the two modalities, dynamically concentrates on various missing regions, and eliminates noise interference. Furthermore, the LEM layer in reconstruction optimization meets the requirements for detailed inpainting and global consistency in complex scenes. The multiangle experiments on three datasets fully demonstrate the generalization ability and reliability of the proposed method.

In the future, we will bridge the gap between synthetic and real datasets to enhance the generalization ability and practical applicability of cloud removal methods. Furthermore, we will explore a multitask collaborative learning framework from extension directions: 1) combining cloud detection to achieve more accurate cloud removal and 2) integrating downstream applications such as land cover classification to improve the generalization ability of the cloud removal model.

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